# HANKEL DETERMINANTS FOR THE $(n+2)$-SHIFTED $q$-METALLIC NUMBERS

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ABSTRACT. For the $q$-deformed metallic number $\Phi_n(q)$ we prove that the Hankel continued fraction of the $(n+2)$-shifted series $\Phi_n^{(n+2)}$ is eventually periodic, and that the corresponding Hankel determinant sequence $\Delta_j^{(n+2)}$ is $2n(n+1)$-(anti)periodic and takes only the values $-2,-1,0,1,2$. This proves part (1) of a conjecture of Han and Pedon. The determinant values arise as telescoping products of the partial numerators of the Hankel continued fraction. The lengthy verifications are carried out by symbolic computation programs, whose code is publicly available on GitHub. We emphasise that these programs give rigorous symbolic proofs, valid for all $n$ and $j$ at once, rather than checks of finitely many cases.

## 1. INTRODUCTION

Morier-Genoud and Ovsienko's theory of $q$-deformed real numbers [3] attaches to the metallic ratio $\phi_n = \frac{1}{2}\big(n+\sqrt{n^2+4}\big)$ a power series $\Phi_n(q) \in \mathbb{Z}[[q]]$, the *$q$-metallic number*. In [2] the Hankel determinants of $\Phi_n$ and of its shifts $\Phi_n^{(\ell)}$ are studied in depth. Write

$$\Delta_j^{(\ell)} := \Delta_j^{(\ell)}(\Phi_n)$$

for the $j$-th Hankel determinant of the $\ell$-shifted series (definitions are recalled in Section 2). For the shifts $0 \le \ell \le n+1$ these determinant sequences are (anti)periodic and take only the values $-1,0,1$; this and the associated Somos–Gale–Robinson dynamics are the main results of [2]. The regime $\ell \ge n+2$ is different, and [2] formulates the following conjecture.

**Conjecture 1** (Han–Pedon [2, Conjecture E])**.**

(1) *The sequence $\Delta_j^{(n+2)}$ is $2n(n+1)$-periodic when $n$ is even and $2n(n+1)$-antiperiodic (hence $4n(n+1)$-periodic) when $n$ is odd, and consists of $-2,-1,0,1,2$ only (except if $n=1$: it can only take the values $-1,0,1$).*

(2) *All sequences $\Delta_j^{(\ell)}$ with $\ell \ge n+3$ are unbounded.*

A Hankel determinant sequence confined to the fixed finite set $\{-2,-1,0,1,2\}$ is a rather unusual phenomenon — all the more striking in view of part (2), where a single further shift already makes the determinants unbounded. The authors have not encountered such behaviour elsewhere.

The purpose of this paper is to prove part (1) of the conjecture. We obtain it from an explicit description of the *Hankel continued fraction* of $\Phi_n^{(n+2)}$ — a continued fraction expansion of a power series, introduced in [1], whose partial numerators determine all the Hankel determinants of the series; see Section 3 for the precise definition.

**Theorem 2.** *Let $n \ge 1$.*

(1) *The Hankel continued fraction of $\Phi_n^{(n+2)}$ is eventually periodic.*

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(2) *The sequence* $\Delta_j^{(n+2)}$ *is* $2n(n+1)$*-periodic when* $n$ *is even and* $2n(n+1)$*-antiperiodic (hence* $4n(n+1)$*-periodic) when* $n$ *is odd, and consists of* $-2,-1,0,1,2$ *only (except if* $n=1$*: it can only take the values* $-1,0,1$*).*

Part (2) is exactly part (1) of Conjecture 1; for $n=1$ it was first proved by Ovsienko and Pedon [4, Theorem 1.6]. We treat all $n \geq 4$ uniformly (the cases $n \leq 3$ are finite computations), and deduce part (2) from part (1).

**Method.** To establish Theorem 2 we use the Hankel continued fraction approach developed in [1]; the underlying idea is recalled in Section 3. Its main tool is an algorithm `NextABC` (Section 3.2), which builds the Hankel continued fraction of a quadratic power series one step at a time. Our strategy proceeds as follows.

1. Derive a quadratic equation $A_0 + B_0F + C_0F^2 = 0$ for the shifted series $F = \Phi_n^{(n+2)}$. This is done in Section 2 (Proposition 4).

2. Apply Algorithm `NextABC` to the initial coefficients $(A_0, B_0, C_0)$ derived above. This produces the terms of a sequence of six-tuples

$$\big(A^{(m+1)}, B^{(m+1)}, C^{(m+1)};\, k^{(m)}, a^{(m)}, D^{(m)}\big), \qquad m = 1, 2, 3, \ldots.$$

From these terms we formulate a closed-form, eventually periodic description of the sequence; see Section 3 (with the full pass-tables in the appendices).

3. Prove that these formulas are correct, i.e. that they satisfy the relations specified in Algorithm `NextABC`. This means checking the three steps of the algorithm. Step 3 (the three relations for the next triple $A^{(m+1)}, B^{(m+1)}, C^{(m+1)}$) is a rational-function identity and involves the longest expressions. Steps 1 and 2 (the leading data $k, a$ and the polynomial $D$) need not be computed at all: by a polynomiality criterion (Lemma 8), they hold as soon as the tuple and the next one are polynomial triples with $B(0) = 1$, $C(0) = 0$, and the displayed $D$ satisfies $D(0) = 1$, $\deg D \leq k+1$ (Section 3.4). Both the Step-3 identities and the polynomiality conditions are verified symbolically — for all $n$ and $j$ at once — by companion Sage programs.

4. Construct the Hankel continued fraction of $F$ from the resulting sequence; see (18) in Section 3.

5. Finally, compute the Hankel determinants $\Delta_j^{(n+2)}$ from the Hankel continued fraction (Section 4). Each nonzero $\Delta_j^{(n+2)}$ is a telescoping product of the partial numerators $v_i = -a^{(i+1)}$. These are constants of the form $\pm 2^e$, and the product collapses to $\pm 1$ or $\pm 2$; all other determinants vanish. This is Theorem 2.

**On the verification.** The computer part of the proof consists of the following Sage programs; each works with $u = q^{2j}$ and $w = q^n$ as free symbols, so that its checks hold for all $n$ and $j$ at once, with no numerical sampling of $n$.

- `passtables.sage` contains the pass-table data, as elements of $\mathbb{Q}(q, u, w)$.
- `certify_step3.sage` proves Step 3 of Algorithm `NextABC` — the relations (14)–(16) — for every transition of the orbit, as identities in $\mathbb{Q}(q, u, w)$. The intermediate expressions are large, exactly as in [1].
- `certify_steps12.sage` proves, by finitely many affine inequalities in $(n, j)$, the polynomiality conditions that give Steps 1 and 2 (Lemma 9).
- `certify_sign.sage` computes the exact sum over one period that decides the sign of the (anti)periodicity in Theorem 2, as a polynomial in $t$, where $n = 2t$ or $n = 2t+1$.
- `verify_base.sage` computes the base cases $n = 1, 2, 3$.

All the programs, their output logs, and the pass-table data are available at

`https://github.com/GuoniuHan/metallicplus`

**Organisation.** Section 2 recalls the shifted series $F = \Phi_n^{(n+2)}$ and derives its initial quadratic equation. Section 3 constructs the Hankel continued fraction of $F$: it introduces the algorithm `NextABC` that generates the tuple sequence, verifies its three steps — Step 3 as a rational-function identity, Steps 1 and 2 by the polynomiality criterion (Lemmas 8 and 9) — and describes the whole orbit (§3.5), proving that it is eventually periodic (Proposition 10). Section 4 proves Theorem 2: the value set via the group lemma (Lemma 12) — along the orbit, the 2-adic exponents of the partial numerators come in a few rigid groups, such as $(1, -2, 1)$ — and the (anti)periodicity via a shift identity for the determinant formula together with two sign facts. The appendices list the explicit pass-tables: the base cases $n = 1, 2, 3$ and, for $n \geq 4$, the two passes and the three bridges, each valid for all $n \geq 4$.

## 2. Setup: the shifted series and its quadratic equation

Throughout, $n \geq 1$ is a fixed integer and

$$Q := q^2 - q + 1, \qquad [n]_q := \frac{1 - q^n}{1 - q} = 1 + q + \cdots + q^{n-1}.$$

We recall from [2] the $q$-deformed metallic number $\Phi_n(q)$, the $q$-deformation of $\phi_n = \frac{n+\sqrt{n^2+4}}{2}$ in the sense of Morier-Genoud and Ovsienko. It is a power series $\Phi_n(q) = \sum_{i \geq 0} f_i q^i \in \mathbb{Z}[[q]]$ with $f_0 = 1$, and it is characterised by the quadratic functional equation

$$q\,\Phi_n(q)^2 + \big((1 + q^n)(1 - q) - q[n]_q\big)\Phi_n(q) = 1. \tag{1}$$

For $\ell \geq 0$, the *$\ell$-shift* of $\Phi_n$ is the power series obtained by deleting its first $\ell$ Taylor coefficients and dividing by $q^\ell$:

$$\Phi_n^{(\ell)}(q) := \frac{1}{q^\ell}\Big(\Phi_n(q) - \sum_{i=0}^{\ell-1} f_i q^i\Big) = \sum_{i \geq 0} f_{i+\ell}\, q^i. \tag{2}$$

For a power series $g(q) = \sum_{i \geq 0} g_i q^i$, the *Hankel determinant* of order $j \geq 1$ is

$$\Delta_j(g) := \det\big(g_{a+b}\big)_{0 \leq a,b \leq j-1}, \tag{3}$$

with the convention $\Delta_0(g) := 1$. The shifted Hankel determinants of $\Phi_n$ are the ordinary Hankel determinants of the shifts, namely $\Delta_j^{(\ell)}(\Phi_n) = \Delta_j\big(\Phi_n^{(\ell)}\big)$; this is the sequence $\Delta_j^{(\ell)}$ studied in the conjecture. Our target is the shift $\ell = n + 2$, i.e. the series $F := \Phi_n^{(n+2)}$.

### 2.1. The quadratic equation of the shift.

The starting point of the whole method is that each shift is again a *quadratic* power series, obtained from the previous one by an elementary substitution.

**Lemma 3** ([2, Lemma 7.1])**.** *Let $F(q)$ be a power series with constant term $c = F(0)$, satisfying $A + BF + CF^2 = 0$ with $A, B, C$ polynomials, $C \neq 0$. Then the shift $F^{(1)}(q) = \dfrac{F(q) - c}{q}$ satisfies $A' + B'F^{(1)} + C'\big(F^{(1)}\big)^2 = 0$, where*

$$A' = \frac{1}{q}\big(A + cB + c^2 C\big), \qquad B' = B + 2cC, \qquad C' = qC. \tag{4}$$

*Proof.* Substitute $F = qF^{(1)} + c$ into $A + BF + CF^2 = 0$:

$$\big(A + cB + c^2 C\big) + q\,(B + 2cC)\,F^{(1)} + q^2 C\,\big(F^{(1)}\big)^2 = 0.$$

The constant term of $A + cB + c^2C$ equals the constant term of the left-hand side of $A + BF + CF^2 = 0$, hence is 0; so the displayed identity is divisible by $q$, giving (4). □

Iterating Lemma 3 from (1) produces the quadratic equation of every shift. Writing $A+BF+CF^2=0$ for (1) in the normalisation $A^{(0)}=-1$, $B^{(0)}=(1+q^n)(1-q)-q[n]_q$, $C^{(0)}=q$, one proves by induction (see [2, Prop. 7.2]) that for $0\le\ell\le n$

$$A^{(\ell)}=\frac{q^{\ell+1}-Q\,(q^n-q^{n-\ell}+1)}{(q-1)^2},\quad B^{(\ell)}=\frac{2q^{\ell+1}-Q\,(q^n+1)}{q-1},\quad C^{(\ell)}=q^{\ell+1},\tag{5}$$

while at $\ell=n+1$

$$A^{(n+1)}=-q^{n-1},\qquad B^{(n+1)}=-\frac{Q+(q^2-3q+1)q^n}{q-1},\qquad C^{(n+1)}=q^{n+2}.\tag{6}$$

The case we need, $\ell=n+2$, is one further step of Lemma 3.

**Proposition 4.** *For every integer $n\ge 2$, the shifted series $F=\Phi_n^{(n+2)}$ is a quadratic power series with integer coefficients, satisfying*

$$A_0+B_0\,F+C_0\,F^2=0,\tag{7}$$

*where*

$$A_0=-q^{n-2},\qquad B_0=-\frac{(q^2-3q+1)\,q^n+Q}{q-1},\qquad C_0=q^{n+3}.\tag{8}$$

*Moreover $B_0,C_0\in\mathbb{Z}[q]$ with $B_0(0)=1$ and $C_0(0)=0$.*

*Proof.* Apply Lemma 3 to (6). The constant term $f_{n+1}=\Phi_n^{(n+1)}(0)$ is determined by evaluating $A^{(n+1)}+B^{(n+1)}f_{n+1}+C^{(n+1)}f_{n+1}^2=0$ at $q=0$: since $n\ge 2$ we have $A^{(n+1)}(0)=0$, $C^{(n+1)}(0)=0$ and $B^{(n+1)}(0)=1$, whence $f_{n+1}=0$. Thus (4) degenerates to

$$A_0=\frac{1}{q}\,A^{(n+1)}=-q^{n-2},\qquad B_0=B^{(n+1)},\qquad C_0=q\,C^{(n+1)}=q^{n+3},$$

which is (8). The numerator of $B_0$ vanishes at $q=1$ ($(1-3+1)+Q(1)=-1+1=0$), so $B_0\in\mathbb{Z}[q]$; a direct check gives $B_0(0)=1$, and $C_0(0)=0$ is clear. □

**Remark 5.** *For $n=1$ the same computation runs one step longer. Formula* (6) *still holds, giving $A^{(2)}=-1$, $B^{(2)}=1+q-q^2$, $C^{(2)}=q^3$, but now the constant term of $\Phi_1^{(2)}$ is $f_2=1$, and Lemma 3 yields for $F=\Phi_1^{(3)}$ the quadratic equation* (7) *with*

$$A_0=Q,\qquad B_0=2q^3-q^2+q+1,\qquad C_0=q^4,\tag{9}$$

*again with $B_0(0)=1$ and $C_0(0)=0$. This small case is treated in Appendix A.*

## 3. The Hankel continued fraction of $\Phi_n^{(n+2)}$

3.1. **Hankel continued fractions.** We first recall from [1] the definition and the basic property of Hankel continued fractions.

**Definition 6.** *For each positive integer $\delta$, a* super continued fraction *associated with $\delta$, called a* super $\delta$-fraction *for short, is a continued fraction of the form*

$$F(q)=\frac{v_0q^{k_0}}{\big|\ 1+u_1(q)q}-\frac{v_1q^{k_0+k_1+\delta}}{\big|\ 1+u_2(q)q}-\frac{v_2q^{k_1+k_2+\delta}}{\big|\ 1+u_3(q)q}-\cdots,\tag{10}$$

*where the $v_j\neq 0$ are constants, the $k_j$ are nonnegative integers, and the $u_j(q)$ are polynomials of degree less than or equal to $k_{j-1}+\delta-2$ (by convention, 0 has degree $-1$).*

When $\delta=1$ (resp. $\delta=2$) and all $k_j=0$, the super $\delta$-fraction (10) is the classical $S$-fraction (resp. $J$-fraction). A super 2-fraction is called a *Hankel continued fraction*, or *$H$-fraction* for short.

**Theorem 7** ([1, Theorem 2.1])**.** (i) *Let $\delta$ be a positive integer. Each super $\delta$-fraction defines a power series, and conversely, for each power series $F(q)$ the super $\delta$-fraction expansion of $F(q)$ exists and is unique.*

(ii) *Let $F(q)$ be a power series whose $H$-fraction is given by* (10) *with $\delta = 2$. Then all non-vanishing Hankel determinants of $F(q)$ are given by*

$$H_{s_j}\big(F(q)\big) = (-1)^{\epsilon_j}\, v_0^{s_j} v_1^{s_j - s_1} v_2^{s_j - s_2} \cdots v_{j-1}^{s_j - s_{j-1}}, \tag{11}$$

*where $\epsilon_j = \sum_{i=0}^{j-1} k_i(k_i+1)/2$ and $s_j = k_0 + k_1 + \cdots + k_{j-1} + j$ for every $j \geq 0$.*

In the statement we have kept the notation of [1]: $H_j(F)$ is the Hankel determinant $\Delta_j(F)$ of (3). We write the $H$-fraction in the standard form

$$\mathcal{H}\big(q; (k_j), (v_j), (u_j)\big) := \cfrac{v_0 q^{k_0}}{1 + u_1(q)q} - \cfrac{v_1 q^{k_0+k_1+2}}{1 + u_2(q)q} - \cfrac{v_2 q^{k_1+k_2+2}}{1 + u_3(q)q} - \cdots . \tag{12}$$

3.2. **The algorithm `NextABC`.** We recall from [1] the reduction underlying the whole computation. Suppose $F(q)$ is a power series satisfying the quadratic equation

$$A(q) + B(q)F(q) + C(q)F(q)^2 = 0.$$

Then $F(q)$ can be expressed in the form

$$F(q) = \frac{-a\, q^k}{D(q) - q^{k+\delta} G(q)},$$

where $G(q)$ is another power series satisfying a transformed quadratic equation

$$A^*(q) + B^*(q)G(q) + C^*(q)G(q)^2 = 0,$$

and the quantities $A^*, B^*, C^*, k, a, D$ can be computed explicitly by means of the following algorithm (here $\delta = 2$), which we also denote by $N$.

**Algorithm** `[NextABC]` (for $\delta = 2$).

Prototype: $(A^*, B^*, C^*;\, k, a, D) = \texttt{NextABC}(A, B, C)$.

Input: $A(q), B(q), C(q) \in \mathbb{Q}[q]$ three polynomials such that $B(0) = 1$, $C(0) = 0$, $C(q) \neq 0$, $A(q) \neq 0$.

Output: $A^*(q), B^*(q), C^*(q) \in \mathbb{Q}[q]$, $k \in \mathbb{N}$, $a \in \mathbb{Q}$ with $a \neq 0$, and $D(q) \in \mathbb{Q}[q]$ a polynomial of degree less than or equal to $k+1$ such that $D(0) = 1$.

Step 1 [Define $k, a$]. Since $A(q) \neq 0$, let $A(q) = aq^k + O(q^{k+1})$ with $a \neq 0$.

Step 2 [Define $D$]. Define $D(q)$ by

$$\frac{aq^k B}{A} - \frac{aq^k C}{B} = D(q) + O(q^{k+2}), \tag{13}$$

where $D(q)$ is a polynomial of degree less than or equal to $k+1$ such that $D(0) = 1$.

Step 3 [Define $A^*, B^*, C^*$]. Let

$$A^*(q) = \big(-D^2 A/a + BD\, q^k - Ca\, q^{2k}\big)/q^{2k+2}, \tag{14}$$

$$B^*(q) = 2AD/(a\, q^k) - B, \tag{15}$$

$$C^*(q) = -A\, q^2/a. \tag{16}$$

*Remarks.* In [1, Algorithm 3.1], $D$ is defined by $-aq^k/F = D - q^{k+2}G$; this is the polynomial of Step 2: from $F = -A/(B + CF)$ we get $-aq^k/F = aq^k B/A + aq^k CF/A$, and $aq^k CF/A = -aq^k C/(B + CF) = -aq^k C/B + O(q^{2k+2})$, since $F = O(q^k)$ and $C = O(q)$. In Step 2, if $C = q^2 C'$ then $aq^k B/A = D(q) + O(q^{k+2})$; similarly, if moreover $A = aq^k$, then $B = D(q) + O(q^{k+2})$. By [1, Lemma 3.2], $A^*, B^*, C^*$ are polynomials.

3.3. **The six-tuple sequence.** Starting from the initial triple $(A_0, B_0, C_0)$ of Proposition 4 and setting $(A^{(1)}, B^{(1)}, C^{(1)}) = (A_0, B_0, C_0)$, repeated application of Algorithm $N = \texttt{NextABC}$ produces six-tuples

$$\big(A^{(m+1)}, B^{(m+1)}, C^{(m+1)};\, k^{(m)}, a^{(m)}, D^{(m)}\big) = N\big(A^{(m)}, B^{(m)}, C^{(m)}\big), \qquad m \geq 1, \tag{17}$$

where $k^{(m)} = \operatorname{val} A^{(m)}$, $a^{(m)}$ is the corresponding leading coefficient (so $K^{(m)} = a^{(m)} q^{k^{(m)}}$), and $D^{(m)}$ is the degree $\leq k^{(m)} + 1$ polynomial of Step 2 of the algorithm. By [1, Lemma 3.2 and Algorithm 3.3] this yields the Hankel continued fraction of $F$,

$$F = \Phi_n^{(n+2)} = \mathcal{H}\big(q; (k_j), (v_j), (u_j)\big), \qquad k_j = k^{(j+1)}, \quad v_j = -a^{(j+1)}, \quad 1 + u_j(q)\, q = D^{(j)} \ \ (j \geq 1). \tag{18}$$

By Theorem 7(ii), the determinants $\Delta_j^{(n+2)}$ are then telescoping products of the $a^{(m)}$.

For $n \geq 2$, the first tuple is

$$\big(A^{(1)}, B^{(1)}, C^{(1)}; K^{(1)}, D^{(1)}\big) = \Big(-q^{n-2},\ B_0,\ q^{n+3};\ -q^{n-2},\ -\tfrac{Q-q^n}{q-1}\Big), \tag{19}$$

with $(A^{(1)}, B^{(1)}, C^{(1)}) = (A_0, B_0, C_0)$ from (8); here $K^{(1)}$ is the leading monomial of $A_0$ (valuation $n-2$, coefficient $-1$) and $D^{(1)} = -(Q - q^n)/(q-1)$ is the polynomial returned by the first application of Algorithm $N$. For $n = 2, 3$ it specialises to $(-1, *, q^5; -1, 1)$ and $(-q, *, q^6; -q, q^2 + 1)$ respectively, matching the base cases of Appendices B and C. For $n \geq 4$ it is the first row of pass 1 (§3.5), and all subsequent tuples of the passes and bridges (Appendices D–H) are generated from it by Algorithm $N$.

3.4. **Verification of one step of Algorithm $N$.** The six-tuples (17) are given explicitly, in closed form, in Appendices A–H; we refer to these as the *pass-tables*, the term "pass" being explained in §3.5. Their entries are rational functions of $q$, $q^{2j}$ and $q^n$. To prove that the tables are correct, we must check that each tuple is sent to the next one by Algorithm $N$, for all $n$ and $j$ in the relevant range.

*Step 3.* The three relations (14)–(16) defining $(A^{(m+1)}, B^{(m+1)}, C^{(m+1)})$ are identities between rational functions, valid whatever the relative size of the exponents. Writing $w = q^n$ and $u = q^{2j}$ (so $q^{4j} = u^2$, $q^{n-2j} = w/u$), they become identities in $\mathbb{Q}(q, u, w)$, valid for all $n$ and $j$ simultaneously. All the transitions occurring below (both parities, the generic and special rows, and the bridges), as well as the junction identities between consecutive passes (§3.5), are verified in this form by the companion program `certify_step3.sage` (which reads the pass-table data from `passtables.sage`).

*Steps 1 and 2.* These steps — the leading data $k, a$ and the truncation (13) defining $D$ — depend on the size of $n$ and $j$, so they cannot be checked as identities. We do not compute them; instead we use the following criterion, which shows that they are forced by polynomiality. Call a triple $(A, B, C)$ *admissible* if

$$A, B, C \in \mathbb{Q}[q], \qquad B(0) = 1, \qquad C(0) = 0,$$

i.e. if it satisfies the input conditions of Algorithm $N$ (apart from $A, C \neq 0$). Every output triple of Algorithm $N$ is admissible: $A^*, B^*, C^*$ are polynomials [1, Lemma 3.2], $C^*(0) = 0$ by (16), and $B^*(0) = 2 \cdot 1 \cdot 1 - 1 = 1$ by (15). The converse holds in the following sharp form.

**Lemma 8** (Polynomiality criterion)**.** *Let $(A, B, C)$ be admissible with $C \neq 0$, and let $k \in \mathbb{N}$, $a \in \mathbb{Q}^*$, and $D \in \mathbb{Q}[q]$ with $D(0) = 1$ and $\deg D \leq k + 1$. Define $A^*, B^*, C^*$ by (14)–(16). Then $(A^*, B^*, C^*;\, k, a, D) = N(A, B, C)$ if and only if $A^*, B^* \in \mathbb{Q}[q]$ and $B^*(0) = 1$.*

*Proof.* The direct implication was noted above. Conversely, by (15), $A/q^k = a(B^* + B)/(2D)$ is a power series, since $D(0) = 1$, and its constant term is $a(1+1)/2 = a$. Hence $A = aq^k + O(q^{k+1})$,

which is Step 1. (In particular $A \neq 0$; indeed, if $A = 0$ then $B^* = -B$ and $B^*(0) = -1$.) Write $A = aq^kU$ with $U(0) = 1$. By (14), $A^* \in \mathbb{Q}[q]$ means

$$\Psi(D) := -UD^2 + BD - aq^kC \equiv 0 \pmod{q^{k+2}}.$$

Let $\widehat{D}$ be the polynomial of Step 2. From (13), $B - U\widehat{D} \equiv aq^kUC/B \pmod{q^{k+2}}$, so $\Psi(\widehat{D}) \equiv aq^kC\,(U\widehat{D}/B - 1)$. The first factor has order at least $k+1$ (recall $C(0) = 0$) and the second has order at least 1 (since $U(0) = \widehat{D}(0) = B(0) = 1$), so $\Psi(\widehat{D}) \equiv 0 \pmod{q^{k+2}}$. Now $\Psi(D) - \Psi(\widehat{D}) = (D - \widehat{D})\big(B - U(D + \widehat{D})\big)$, and the second factor takes the value $1 - 2 = -1$ at $q = 0$. Hence $D \equiv \widehat{D} \pmod{q^{k+2}}$, and $D = \widehat{D}$ since both have degree at most $k+1$. Then Step 3 gives $A^*, B^*, C^*$. □

We apply the lemma to a tuple $(A, B, C; K, D)$ of the pass-tables, $K = aq^k$, at a given $(n, j)$. By Step 3 (verified as an identity), the triple $(A^*, B^*, C^*)$ it produces is the triple of the next tuple, and $C^* = -Aq^2/a$ is a polynomial with $C^*(0) = 0$ as soon as $A$ is a polynomial. Hence:

> *the $k, a, D$ written in a tuple are those of Algorithm N exactly when the tuple and the next one are both admissible,*

provided $k \geq 0$, $D(0) = 1$ and $\deg D \leq k + 1$. In particular, when the next tuple is *not* admissible, the $k, a, D$ of the tuple are wrong; this explains why a pass ends (§3.5).

**Lemma 9** (Steps 1 and 2)**.** *For every tuple in the pass-tables and every $(n, j)$ at which it occurs in the sequence described in §3.5, the tuple and the next one are admissible, and $k \geq 0$, $D(0) = 1$, $\deg D \leq k + 1$. Consequently $k^{(m)}, a^{(m)}, D^{(m)}$ are as displayed.*

*Proof.* All conditions are statements about $q$-orders. Write each entry as a finite sum of *blocks*

$$X = \sum_{b,c} P_{b,c}(q)\, u^b w^c, \qquad u = q^{2j},\ w = q^n,\ P_{b,c} \in \mathbb{Q}(q).$$

For the pass-table entries, $(b, c)$ ranges over $(0,0), (1,0), (-1,1), (0,1)$ — the blocks $1, q^{2j}, q^{n-2j}, q^n$ — and the denominator of $P_{b,c}$ is $\lambda\, q^s (q-1)^e$ with $\lambda \in \mathbb{Q}^*$. Under $u = q^{2j}$, $w = q^n$ the block $(b, c)$ has $q$-order

$$\operatorname{ord}(b, c) = \operatorname{val}_q P_{b,c} + 2jb + nc, \tag{20}$$

an affine form in $(j, n)$. An entry $X$ is a polynomial in $q$ as soon as (a) the factor $(q-1)^e$ of its denominator divides the numerator after the substitution $u = q^{2j}$, $w = q^n$, and (b) every block has $\operatorname{ord}(b, c) \geq 0$. Condition (a) is checked identically in $j$ and $n$, by expanding the numerator at $q = 1 + \varepsilon$ with $j, n$ symbolic (the coefficients of $\varepsilon^0, \ldots, \varepsilon^{e-1}$ are polynomials in $j, n$, and they vanish). Conditions such as (b), and likewise $B(0) = 1$ (the block 1 of $B$ starts with 1 and every other block has order $\geq 1$), $C(0) = 0$, $k \geq 0$, $D(0) = 1$ and $\deg D \leq k + 1$, are inequalities between affine forms in $(j, n)$, to be proved on a *cone* $\mathcal{C} = \{\, j \geq j_0,\ n - 2j \geq d_0 \,\}$. For an affine form $g = \alpha + \beta j + \gamma n$ and a target $\tau \in \mathbb{Z}$, substituting $j = j_0 + s$ and $n - 2j = d_0 + s'$ with $s, s' \geq 0$ gives the finite test

$$g(j, n) \geq \tau \ \text{ on } \mathcal{C} \iff \beta + 2\gamma \geq 0, \quad \gamma \geq 0, \quad \alpha + \beta j_0 + \gamma(2j_0 + d_0) \geq \tau. \tag{21}$$

The companion program `certify_steps12.sage` walks the pass-tables in orbit order, with their case conditions (generic rows, special rows $j = 0$, bridges). For each tuple it computes the image $(A^*, B^*)$ by Step 3, proves on a cone the three facts (i) the tuple is admissible, (ii) $k \geq 0$, $D(0) = 1$, $\deg D \leq k + 1$, (iii) $A^*, B^*$ are polynomials with $B^*(0) = 1$, and checks *coverage*: the orbit uses the tuple only inside that cone. For a generic row, the least $n - 2j$ at which it is used is a constant read off from the exit index of the pass (§3.5); the $j$-free tuples are used for all $n \geq 4$. All tuples pass. No value of $n$ or $j$ is ever substituted. □

*Worked instance.* Take $m = 9j + 3$ in pass 1 with $j \geq 1$, and write $d = n - 2j$. In blocks, the pass-table (Appendix D) reads

$$A = 2q^{d-3}, \qquad K = 2q^{d-3}, \qquad B = -\frac{Q}{q-1}\,(1+q^n) - \frac{2(q-2)}{q(q-1)}\,q^d,$$
$$C = \frac{Qq^2(q-2)}{2(q-1)^2}\,(1+q^n) + \frac{q(q-2)^2}{2(q-1)^2}\,q^d + \frac{Qq^4}{2(q-1)^2}\,q^{2j}, \qquad D = \frac{q^{d-1}-Q}{q-1} = [d-1]_q - q.$$

(i) The orders of the blocks of $A$ are $d-3$; of $B$, 0 (with $-Q/(q-1) = 1 + O(q)$), $n$ and $d-1$; of $C$, 2, $n+2$, $d+1$ and $2j+4$. With the cancellation of $(q-1)^e$, the triple is admissible for $d \geq 3$.
(ii) $k = d-3 \geq 0$, $D(0) = 1$ and $\deg D \leq d-2 = k+1$, with equality for $d \geq 4$ (for $d = 3$, $D = 1$).
(iii) Step 3 gives

$$A^* = -Q\,[n]_q - q^{d-1} + Q\,q^{2j+1}, \qquad B^* = Q\,[n]_q + 2q^{d-1},$$

which are the $A, B$ of row $m = 9j + 4$; they are polynomials, and $B^*(0) = 1$ for $d \geq 2$. Hence the row is correct on $\{j \geq 1,\ d \geq 3\}$, and the orbit uses it only there. At $d = 2$ the row itself is not admissible ($A = 2q^{-1}$); this is why, for even $n$, pass 1 stops at row 2 of its last block (§3.5).

3.5. **The orbit.** We write $m = 9j + r$ with $r \in \{1, \ldots, 9\}$: block $j \geq 0$ consists of $m = 9j + 1, \ldots, 9j + 9$, and block 0 starts with the initial tuple $m = 1$. Put $d := n - 2j$. A *pass* is a run of blocks given by one family of nine formulas: pass 1 (Appendix D) and pass 2 (Appendix E); along a pass, $d$ decreases by 2 from one block to the next. The passes are linked by three *bridges*, free of $j$ (Appendices F–H). All these formulas are valid for every $n \geq 4$. A *row* is one of the nine formulas of a pass, and a *tuple* is the value of a row (or of a bridge entry) at a given $(n, j)$. The rows of pass 1 at $j = 0$ for $m \leq 6$ are given separately in Appendix D (the *special rows*); all other rows are *generic*. The *exit* of a pass is the place where the orbit leaves it: the first row whose successor is not admissible.

Items (a)–(f) below describe a sequence of tuples $x_1, x_2, \ldots$ taken from the tables, with $x_1$ the initial tuple (19). Each step $x_m \to x_{m+1}$ is a transition of Algorithm $N$, by Step 3 (§3.4) and Lemma 9; by induction on $m$, the sequence $(x_m)$ is therefore the orbit, and we only have to say which tuple comes next. By Lemma 8, the $K, D$ of a row are correct exactly where the row and the next one are both admissible. We write $p_r$ (pass 1) and $p'_r$ (pass 2) for the bound in the header $n - 2j \geq p_r$ of row $r$ in Appendices D and E, on which the row is admissible: $p = (0, 1, 3, 2, 0, 0, 4, 3, 1)$ and $p' = (0, 3, 2, 0, 0, 2, 4, 3, 1)$. Since all these bounds are at most 4, the orbit follows a pass through every block with $d \geq 4$, and into the first row of the next block; the last block is examined below. (Strictly speaking, the exits are not needed for the proof: once a bridge tuple has the right $A, B, C$ and satisfies Lemma 8, it is the next tuple, since Algorithm $N$ is deterministic. The exits explain why the pass cannot continue.)

*The orbit for even $n \geq 4$.*

(a) *Start.* The initial tuple (19) is row 1 of pass 1 at $j = 0$. Pass 1 runs through the blocks $j = 0, 1, \ldots$, i.e. $d = n, n-2, \ldots$, with the special rows $j = 0$ for $m = 1, \ldots, 6$ and the generic rows otherwise.

(b) *Exit of pass* 1. All blocks with $d \geq 4$ are run through. The last block has $d = 2$, $j = \frac{n}{2} - 1$. Rows 1 and 2 are admissible there ($p_1 = 0$, $p_2 = 1$), so the step from row 1 is correct and the next tuple has the $A, B, C$ of row 2. But row 3 is not admissible ($A = 2q^{d-3} = 2q^{-1}$), so the $K, D$ of row 2 are wrong: the tuple at the index

$$M := 9\Big(\frac{n}{2} - 1\Big) + 2 = \frac{9}{2}n - 7$$

is not given by row 2, and pass 1 occupies $m < M$.

(c) *Bridge* 2. The tuple at $m = M$ is c1: it has the $A, B, C$ of row 2 at $q^{2j} = q^{n-2}$ (an identity in $\mathbb{Q}(q, w)$, proved in `certify_step3.sage`), with its own $K, D$. The orbit runs through $\mathrm{c1}, \ldots, \mathrm{c5}$, and c5 is sent to row 6 of pass 2 at $j = 0$.

(d) *Pass* 2. It is read at integer $j$, so $d = n, n-2, \ldots$ is even, and runs through the rows $9j + r$ of Appendix E in increasing order, from $9j + r = 6$ to $\frac{9}{2}n - 9$ (row 9 at $j = \frac{n}{2} - 2$, $d = 4$). In the next block, $d = 2$, row 1 is admissible ($p'_1 = 0$) but row 2 is not ($A = Q\,q^{d-3} = Q\,q^{-1}$), so the tuple at the position of row 1 of that block is not given by row 1: pass 2 ends just before it.

(e) *Bridge* 1. The tuple at that position is b1: it has the $A, B, C$ of row 1 of pass 2 at $q^{2j} = q^{n-2}$ (proved in `certify_step3.sage`), with $K = q$, $D = Q$ instead of $D = 1 - q$. The orbit runs through $\mathrm{b1}, \ldots, \mathrm{b8}$, and b8 is sent back to pass 1, to its tuple $m = 2$ (row 2 at $j = 0$), which is the second tuple of the orbit.

(f) *Period.* Since $\mathrm{b8} = (q^{n-2}, B^{(1)}, -q^{n+3}) = (-A^{(1)}, B^{(1)}, -C^{(1)})$ differs from the initial tuple, and both are sent to the tuple $m = 2$ of pass 1, the orbit is periodic from $m = 2$ on, with pre-period exactly 1. One period consists of $M - 2 = \frac{9}{2}n - 9$ tuples of pass 1 ($m = 2, \ldots, M-1$), 5 of bridge 2, $\frac{9}{2}n - 14$ of pass 2 and 8 of bridge 1: in total $9n - 10$.

*The orbit for odd $n \geq 5$.*

(a) *Start.* As for even $n$: the initial tuple is row 1 of pass 1 at $j = 0$, and pass 1 runs through $d = n, n-2, \ldots$, now odd.

(b) *Exit of pass* 1. All blocks with $d \geq 5$ are run through. The last block has $d = 3$, $j = \frac{n-3}{2}$. Rows 1–6 are admissible there ($p_1, \ldots, p_6 \leq 3$), so the steps from rows 1–5 are correct and the tuple at row 6 has the $A, B, C$ of row 6. But row 7 is not admissible ($A = q^{d-4} = q^{-1}$), so the $K, D$ of row 6 are wrong: the tuple at the index

$$M := 9 \cdot \frac{n-3}{2} + 6 = \frac{9n-15}{2}$$

is not given by row 6, and pass 1 occupies $m < M$.

(c) *Bridge* 3. The tuple at $m = M$ is d1: it has the $A, B, C$ of row 6 at $q^{2j} = q^{n-3}$ (proved in `certify_step3.sage`), with its own $K, D$. The orbit runs through $\mathrm{d1}, \mathrm{d2}, \mathrm{d3}$, and d3 is sent to row 2 of pass 2 at $j = \frac{1}{2}$.

(d) *Pass* 2. It is read at half-integer $j \in \mathbb{Z} + \frac{1}{2}$, so again $d = n-1, n-3, \ldots$ is even, and runs through the rows $9j + r$ of Appendix E in increasing order (this index is local to pass 2, and here a half-integer), from $9j + r = \frac{13}{2}$ to $\frac{9}{2}n - 9$ (row 9 at $j = \frac{n}{2} - 2$, $d = 4$). In the next block, $d = 2$, it ends exactly as for even $n$.

(e) *Bridge* 1. As for even $n$: the next tuple is b1, the orbit runs through $\mathrm{b1}, \ldots, \mathrm{b8}$, and b8 is sent back to the tuple $m = 2$ of pass 1.

(f) *Period.* As for even $n$, the pre-period is exactly 1. One period consists of

$$M - 2 = \tfrac{9n-19}{2} \text{ (pass 1)}, \quad 3 \text{ (bridge 3)}, \quad \tfrac{9n-29}{2} \text{ (pass 2)}, \quad 8 \text{ (bridge 1)}$$

tuples, in total $9n - 13$.

*The cases $n = 1, 2, 3$.* Since Algorithm $N$ runs verbatim at a numerical $n$, the tuples are not guessed but computed by iterating it from the initial triple, in the program `verify_base.sage`; they are listed in Appendices A–C, and are periodic from $m = 2$ on, with periods 2, 9 and 14.

*Why this is not automatic.* The tuples range over an infinite set, so — unlike the situation over a finite field, where every orbit is eventually periodic by pigeonhole [1] — there is no general reason for the orbit to close up; it is a fact established by carrying out Algorithm $N$ explicitly, as above.

**Proposition 10.** *For every $n \geq 1$, the six-tuple sequence* (17) *is eventually periodic, with pre-period* 1 *and period*

$$P_n = \begin{cases} 2,\ 9,\ 14 & \text{for } n = 1, 2, 3, \\ 9n - 10 & \text{for even } n \geq 4, \\ 9n - 13 & \text{for odd } n \geq 5. \end{cases}$$

Here $P_n$ is *a* period; we do not need it to be the minimal one. The pre-period is exactly 1 in any case: if $x_1$ recurred, it would recur at index $1 + P_n$, but $x_{1+P_n} = (-A^{(1)}, B^{(1)}, -C^{(1)}) \neq x_1$.

**Remark 11.** *Reading the values of $K$ in the tables along the orbit, every partial numerator $v_j = -a^{(j+1)}$ is a nonzero constant of the form $\pm 2^e$ with $e \in \{-2, -1, 0, 1, 2\}$.*

Proposition 10 and Remark 11 reduce Theorem 2 to a computation with the data $(k_j, v_j)$ over one period: these are given by finitely many affine formulas, and we must show that the resulting Hankel determinants (23) lie in $\{-2, -1, 0, 1, 2\}$ and are (anti)periodic. This is carried out in the next section.

## 4. Proof of Theorem 2

We apply Theorem 7(ii) to the $H$-fraction (18) of $F = \Phi_n^{(n+2)}$. With the data $(k_j, v_j)$ of (18), formula (11) says that the only nonzero Hankel determinants of $F$ are the $\Delta_{s_j}(F)$ at the indices

$$s_0 = 0, \qquad s_j = k_0 + k_1 + \cdots + k_{j-1} + j \quad (j \geq 1), \tag{22}$$

and there

$$\Delta_{s_j}(F) = (-1)^{\epsilon_j}\, v_0^{s_j}\, v_1^{s_j - s_1} \cdots v_{j-1}^{s_j - s_{j-1}}, \qquad \epsilon_j = \sum_{i=0}^{j-1} \binom{k_i + 1}{2}. \tag{23}$$

All other Hankel determinants of $F$ vanish; these account for the value 0 in Theorem 2. It therefore remains to show that each product (23) equals $\pm 1$ or $\pm 2$.

4.1. **Reduction to an exponent count.** By Remark 11 every partial numerator is of the form $v_i = \pm 2^{e_i}$ with $e_i \in \mathbb{Z}$. Writing $\sigma_i = \operatorname{sgn} v_i$, the product (23) is

$$\Delta_{s_j}(F) = (-1)^{\epsilon_j} \left( \textstyle\prod_{i<j} \sigma_i^{\,s_j - s_i} \right) 2^{E_j}, \qquad E_j := \textstyle\sum_{i=0}^{j-1} e_i\,(s_j - s_i). \tag{24}$$

Thus $\Delta_{s_j}(F) \in \{\pm 1, \pm 2\}$ if and only if $E_j \in \{0, 1\}$, whatever the sign. So the value-set part of Theorem 2 is equivalent to the purely arithmetic statement

$$E_j \in \{0, 1\} \qquad \text{for all } j \geq 0. \tag{25}$$

4.2. **The group lemma.** Recall $s_{i+1} - s_i = k_i + 1$, and write

$$\kappa_i := k_i + 1 \geq 1, \qquad T_i := \sum_{l=0}^{i} e_l, \tag{26}$$

so that $T_i$ is the $i$-th partial sum of the exponent sequence. Reversing the order of summation in (24),

$$E_j = \sum_{i=0}^{j-1} e_i \sum_{l=i}^{j-1} \kappa_l = \sum_{l=0}^{j-1} \kappa_l \sum_{i=0}^{l} e_i = \sum_{l=0}^{j-1} \kappa_l\, T_l, \qquad \text{i.e.} \qquad E_{j+1} = E_j + \kappa_j\, T_j. \tag{27}$$

So $E$ is driven by the running total $T$, and only the tuples with $T_j \neq 0$ move it. The exponents $e_i$ come in a few rigid groups. Call a *group* one of the words

$$(0), \qquad G_3 := (1, -2, 1), \qquad G_8 := (1, -2, 2, -1, -1, 2, -2, 1), \qquad G_7 := (1, -2, 2, -2, 2, -2, 1).$$

**Lemma 12.** *Let $n \geq 2$. The exponent sequence $(e_i)_{i\geq 0}$ of the H-fraction of $\Phi_n^{(n+2)}$ is a concatenation of groups: (0) and $G_3$, together with one $G_8$ per period when $n \geq 4$ is even; for $n = 2$ the groups are (0) and one $G_7$ per period. Moreover $k_i = 0$ at every letter where the running sum of the letters of the group is nonzero, namely at letters $1, 2$ of $G_3$, letters $1, 2, 3, 5, 6, 7$ of $G_8$, and letters $1, \ldots, 6$ of $G_7$.*

*Proof.* For $n = 2, 3$ the orbit is the finite list of Appendices B and C. For $n = 2$ the exponents over one period ($m = 2, \ldots, 10$) are $G_7, 0, 0$, with $k = 0$ on $G_7$; for $n = 3$ they are $G_3$ followed by zeros and a second $G_3$, with $k = 0$ at the required letters.

Let $n \geq 4$. We follow the orbit of §3.5 and read $a^{(m)}$, hence $e = \nu_2(a^{(m)})$ (the 2-adic valuation), and $k^{(m)}$ from the values of $K$ in the tables. In every tuple $K = a\,q^k$ with $a$ a constant, independent of $n$ and $j$.

*Generic rows.* For the generic rows of a pass, $a$ depends only on the residue $r$ (for pass 2, $r$ is the residue in the index $9j + r$ of Appendix E), and $k$ is affine in $j$ and $d = n - 2j$:

| $r$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| $a$, pass 1 | 2 | $-\frac{1}{4}$ | 2 | $-1$ | 1 | 1 | 1 | 1 | $-1$ |
| $k$, pass 1 | 0 | 0 | $d-3$ | 0 | $2j-1$ | 1 | $d-4$ | 0 | $2j$ |
| $e$, pass 1 | 1 | $-2$ | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| $a$, pass 2 | 1 | 1 | 1 | $-1$ | 2 | $-\frac{1}{4}$ | 2 | $-1$ | 1 |
| $k$, pass 2 | 1 | $d-3$ | 0 | $2j-1$ | 0 | 0 | $d-4$ | 0 | $2j$ |
| $e$, pass 2 | 0 | 0 | 0 | 0 | 1 | $-2$ | 1 | 0 | 0 |

So a full block of pass 1 reads $G_3, 0, \ldots, 0$, a full block of pass 2 reads $0, 0, 0, 0, G_3, 0, 0$, and in both cases $k = 0$ at the first two letters of $G_3$.

*The other tuples.* The rows $m = 1, \ldots, 6$ of pass 1 at $j = 0$ and the bridges have

| | $m$=1 | 2 | 3 | 4 | 5 | 6 | c1 | c2 | c3 | c4 | c5 | d1 | d2 | d3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $a$ | $-1$ | 2 | $\frac{1}{4}$ | 2 | $-1$ | $-1$ | $-\frac{1}{4}$ | 4 | $\frac{1}{2}$ | $\frac{1}{2}$ | 4 | 1 | $-1$ | $-1$ |
| $k$ | $n-2$ | 0 | 0 | $n-3$ | 0 | 1 | 0 | 0 | $n-3$ | 0 | 0 | 1 | $n-1$ | 1 |
| $e$ | 0 | 1 | $-2$ | 1 | 0 | 0 | $-2$ | 2 | $-1$ | $-1$ | 2 | 0 | 0 | 0 |

| | b1 | b2 | b3 | b4 | b5 | b6 | b7 | b8 |
|---|---|---|---|---|---|---|---|---|
| $a$ | 1 | $-1$ | $-1$ | 2 | $\frac{1}{4}$ | 2 | 1 | 1 |
| $k$ | 1 | 0 | $n-3$ | 0 | 0 | $n-2$ | $n+1$ | $n-2$ |
| $e$ | 0 | 0 | 0 | 1 | $-2$ | 1 | 0 | 0 |

*Along the orbit.* We now cut the orbit of §3.5 into groups.

- The pre-period $m = 1$ gives (0), and the rows $m = 2, \ldots, 6$ give $G_3, 0, 0$, with $k = 0$ at $m = 2, 3$. The rows $7, 8, 9$ at $j = 0$ and the following full blocks of pass 1 give zeros and copies of $G_3$, as above.
- *$n$ even.* The last block of pass 1 ($d = 2$) contributes only row 1; then come c1, …, c5 and rows 6, 7 of pass 2 at $j = 0$. Their exponents are $1, -2, 2, -1, -1, 2, -2, 1$, which is $G_8$, and their $k$ are $0, 0, 0, n-3, 0, 0, 0, n-4$: zero at letters $1, 2, 3, 5, 6, 7$. Pass 2 continues with rows 8, 9 (zeros) and full blocks.
- *$n$ odd.* The last block of pass 1 ($d = 3$) contributes rows 1–5, i.e. $G_3, 0, 0$, with $k = 0$ at rows 1, 2. Then d1, d2, d3 give zeros, and pass 2 is entered at row 2, so rows $2, 3, 4$ give zeros and the block continues as above.
- *Both parities.* Pass 2 is left after row 9 of its last full block. Then b1, …, b8 give $0, 0, 0, G_3, 0, 0$, with $k = 0$ at b4, b5, and the orbit returns to $m = 2$.

This covers one period, and the lemma follows. □

**Corollary 13.** *For every $n \geq 1$ and every $j \geq 0$, $E_j \in \{0,1\}$; moreover $E_j = 0$ when $e_0, \dots, e_{j-1}$ is a concatenation of whole groups.*

*Proof.* For $n = 1$ all partial numerators are $\pm 1$ (Appendix A), so $e_i \equiv 0$ and $E_j \equiv 0$. Let $n \geq 2$. Every group has exponent sum 0, so the running total $T$ is 0 at the end of every group, and by (27) $E$ moves only inside a group. Starting a group with $E = 0$, and using $\kappa_i = 1$ wherever Lemma 12 gives $k_i = 0$, the values of $T$ and of $E$ along each group are

| group | $T$ | $E$ |
|---|---|---|
| (0) | $0$ | $0, 0$ |
| $G_3$ | $1, -1, 0$ | $0, 1, 0, 0$ |
| $G_8$ | $1, -1, 1, 0, -1, 1, -1, 0$ | $0, 1, 0, 1, 1, 0, 1, 0, 0$ |
| $G_7$ | $1, -1, 1, -1, 1, -1, 0$ | $0, 1, 0, 1, 0, 1, 0, 0$ |

(at the letters where $T = 0$ the value of $k$ does not matter). Hence $E$ is 0 at the end of every group and $E_j \in \{0,1\}$ for all $j$. □

4.3. **The (anti)periodicity.** Put $L := 2n(n+1)$ and let $P = P_n$ be the period of Proposition 10.

**Lemma 14** (Shift lemma)**.** *Assume that*

$$k_{i+P} = k_i \quad (i \geq 0), \qquad v_{i+P} = v_i \quad (i \geq 1), \qquad v_P = -v_0, \qquad s_P = L.$$

*Then for every $j \geq 0$*

$$\Delta_{s_{j+P}}(F) = \Big(-\prod_{i<P} v_i\Big)^{s_j} \Delta_L(F)\, \Delta_{s_j}(F). \tag{28}$$

*If moreover $\prod_{i<P} v_i = -1$, then $\Delta_{h+L}(F) = \Delta_L(F)\, \Delta_h(F)$ for every $h \geq 0$.*

*Proof.* The assumptions give $s_{P+i} = s_i + L$ for all $i \geq 0$, and $\epsilon_{j+P} = \epsilon_P + \epsilon_j$. Split the product (23) for $\Delta_{s_{j+P}}$ into the factors $i < P$ and $i = P + i'$ with $i' < j$. The first part is $\prod_{i<P} v_i^{\,s_{j+L-s_i}} = \big(\prod_{i<P} v_i\big)^{s_j} \prod_{i<P} v_i^{\,s_P - s_i}$. In the second part, $v_{P+i'}^{\,s_{j+P}-s_{P+i'}} = v_{P+i'}^{\,s_j - s_{i'}}$, which is $v_{i'}^{\,s_j - s_{i'}}$ for $i' \geq 1$ and $(-1)^{s_j} v_0^{\,s_j}$ for $i' = 0$. Collecting the signs $(-1)^{\epsilon_{j+P}} = (-1)^{\epsilon_P}(-1)^{\epsilon_j}$ gives (28), since $\Delta_L = \Delta_{s_P}$. If $\prod_{i<P} v_i = -1$, then (28) reads $\Delta_{s_j+L} = \Delta_L \Delta_{s_j}$. Finally, since $s_{j+P} = s_j + L$, an index $h \geq 0$ is of the form $s_j$ if and only if $h + L$ is, and the other indices give 0 on both sides. □

*The hypotheses hold for our orbit.* The tuple at index $1 + P$ is $\big(-A^{(1)}, B^{(1)}, -C^{(1)}\big)$ — the initial tuple up to the signs of $A$ and $C$ (it is b8 for $n \geq 4$, and the last listed tuple for $n = 1, 2, 3$). Since the orbit is periodic from $m = 2$ on, this gives the first three conditions, in terms of (18). For the last one, $s_P = \sum_{i<P}(k_i + 1)$; for $n \geq 4$, summing the values of $k$ in the tables over $m = 1, \dots, P$ segment by segment (one full block of either pass contributes $9 + (2n - 7) = 2n + 2$, by the $k$-table in the proof of Lemma 12) gives

| segment | $n$ even | $n$ odd |
|---|---|---|
| $m = 1, \dots, 6$ $(j = 0)$ | $2n + 2$ | $2n + 2$ |
| pass 1, rows 7, 8, 9 at $j = 0$ | $n - 1$ | $n - 1$ |
| full blocks of pass 1 | $(\frac{n}{2} - 2)(2n + 2)$ | $\frac{n-5}{2}(2n + 2)$ |
| last block of pass 1 | 1 (row 1) | $n + 1$ (rows 1–5) |
| bridge 2, resp. 3 | $n + 2$ | $n + 4$ |
| pass 2 before its full blocks | $n$ (rows 6–9, $j = 0$) | $2n$ (rows 2–9, $j = \frac{1}{2}$) |
| full blocks of pass 2 | $(\frac{n}{2} - 2)(2n + 2)$ | $\frac{n-5}{2}(2n + 2)$ |
| b1, …, b7 | $3n + 4$ | $3n + 4$ |
| total | $2n(n + 1)$ | $2n(n + 1)$ |

For $n = 1, 2, 3$ this is a sum over the tables of Appendices A–C.

**Lemma 15** (Two sign facts)**.** *For every $n \geq 1$:* (I) $\prod_{i<P} v_i = -1$*;* (II) $\Delta_L(F) = (-1)^n$.

*Proof.* (I) By Lemma 12 (for $n = 1$ all $e_i = 0$) the exponents $e_0, \ldots, e_{P-1}$ form whole groups, so their sum is 0 and $\left|\prod_{i<P} v_i\right| = 1$. As $v_i = -a^{(i+1)}$, the sign is $(-1)^P \prod_{m\leq P} \operatorname{sgn} a^{(m)} = (-1)^{\#\{m\leq P:\, a^{(m)}>0\}}$. For $n \geq 4$ we count the positive $a^{(m)}$ among the values of $K$, $m = 1, \ldots, P$: a full block of either pass has 6 of them, and the remaining segments give

| | $m \leq 6$ | rows 7–9 | last block | bridge | pass 2 start | b1–b7 | total |
|---|---|---|---|---|---|---|---|
| $n$ even | 3 | 2 | 1 | 4 | 2 | 5 | 17 |
| $n$ odd | 3 | 2 | 3 | 1 | 5 | 5 | 19 |

which is odd in both cases. For $n = 1, 2, 3$ it is read off Appendices A–C (and checked in `verify_base.sage`).

(II) By (24) with $j = P$, and $E_P = 0$ (Corollary 13), we have $|\Delta_L(F)| = 1$, and the sign of $\Delta_L(F)$ is $(-1)^{\epsilon_P}$ times $(-1)^{L-s_i}$ for each $i < P$ with $v_i < 0$. Since $L$ is even, $\Delta_L(F) = (-1)^{X(n)}$ with

$$X(n) := \epsilon_P + \sum_{i<P,\ v_i<0} s_i.$$

For $n \geq 4$, write $n = 2t$ or $n = 2t + 1$. Along one period every $k^{(m)}$ is an affine function of $t$ and of the block number $j$ (read off the values of $K$), so every $s_i$ is a polynomial in $t$ and $j$, and $X(n)$ is obtained by exact summation over the segments above. The companion program `certify_sign.sage` carries this out symbolically and finds

$$X(2t) = \tfrac{160}{3}t^3 - 24t^2 + \tfrac{2}{3}t, \qquad X(2t+1) = \tfrac{160}{3}t^3 + 40t^2 + \tfrac{2}{3}t - 1.$$

In both cases $(X(n) - n)/2$ is a polynomial in $t$ of degree 3 taking integer values at $t = 0, 1, 2, 3$, hence at every integer (Newton's forward-difference formula); so $X(n) \equiv n \pmod 2$. For $n = 1, 2, 3$, (II) is a finite computation (`verify_base.sage`). □

*Proof of Theorem 2. Part* (1) is Proposition 10: for every $n$ the six-tuple sequence, hence the $H$-fraction data $(k_j, v_j, u_j)$ of (18), is eventually periodic.

*Part* (2)*, the value set.* By (24) and Corollary 13, every nonzero $\Delta_j^{(n+2)}$ equals $\pm 2^{E_j} \in \{\pm 1, \pm 2\}$, while the remaining indices give $\Delta_j^{(n+2)} = 0$. For $n = 1$, $E_j \equiv 0$ and $\Delta_j^{(3)} \in \{-1, 0, 1\}$, recovering [4, Theorem 1.6].

*Part* (2)*, the (anti)periodicity.* By Lemma 14 (whose hypotheses hold for our orbit, see §4.3) and Lemma 15, $\Delta_{h+2n(n+1)}^{(n+2)} = (-1)^n \Delta_h^{(n+2)}$ for all $h \geq 0$: the sequence is $2n(n+1)$-periodic for $n$ even and $2n(n+1)$-antiperiodic for $n$ odd. □

## Appendix A. Base case $n = 1$

The initial triple is given by Remark 5; write $B := 2q^3 - q^2 + q + 1$, so $B^{(m)} = B$ for all $m$. The orbit has pre-period 1 and period 2 (from $m = 2$), and $m = 3$ maps back to $m = 2$; the three transitions are identities between concrete polynomials in $q$, computed in `verify_base.sage`. Note that all $a^{(m)} = \pm 1$ and that $\sum_m \left(k^{(m)} + 1\right) = 3 + 1 = 4 = 2n(n+1)$ over one period.

$m = 1$:

$$A^{(m)} = Q, \qquad C^{(m)} = q^4, \qquad K^{(m)} = 1, \qquad D^{(m)} = 2\,q + 1$$

$m = 2$:

$$A^{(m)} = -q^2, \qquad C^{(m)} = -Qq^2, \qquad K^{(m)} = -q^2, \qquad D^{(m)} = B$$

$m = 3$:

$$A^{(m)} = -Q, \qquad C^{(m)} = -q^4, \qquad K^{(m)} = -1, \qquad D^{(m)} = 2\,q + 1$$

## Appendix B. Base case $n = 2$

This appendix lists the 10 tuples $\big(A^{(m)}, B^{(m)}, C^{(m)}; K^{(m)}, D^{(m)}\big)$ for $m = 1, \dots, 10$, computed by `verify_base.sage`. The period is 9 (from $m = 2$), and $m = 10$ maps back to $m = 2$. $m = 1$:

$$A^{(m)} = -1, \qquad B^{(m)} = -q^3 + 2\,q^2 + 1, \qquad C^{(m)} = q^5, \qquad K^{(m)} = -1, \qquad D^{(m)} = 1$$

$m = 2$:

$$A^{(m)} = q^3 - q + 2, \qquad B^{(m)} = q^3 - 2\,q^2 + 1, \qquad C^{(m)} = -q^2, \qquad K^{(m)} = 2$$
$$D^{(m)} = \frac{1}{2}\,q + 1$$

$m = 3$:

$$A^{(m)} = -\frac{1}{8}\,q^3 - \frac{3}{8}\,q + \frac{1}{4}, \qquad B^{(m)} = \frac{1}{2}\,q^4 + \frac{3}{2}\,q^2 + 1, \qquad C^{(m)} = -\frac{1}{2}\,q^5 + \frac{1}{2}\,q^3 - q^2$$
$$K^{(m)} = \frac{1}{4}, \qquad D^{(m)} = \frac{3}{2}\,q + 1$$

$m = 4$:

$$A^{(m)} = 2\,q^3 + 2\,q^2 + 6\,q + 4, \qquad B^{(m)} = -2\,q^4 - q^3 - 6\,q^2 + 1, \qquad C^{(m)} = \frac{1}{2}\,q^5 + \frac{3}{2}\,q^3 - q^2$$
$$K^{(m)} = 4, \qquad D^{(m)} = -\frac{3}{2}\,q + 1$$

$m = 5$:

$$A^{(m)} = -\frac{1}{8}\,q^3 - \frac{1}{8}\,q^2 - \frac{3}{8}\,q - \frac{1}{4}, \qquad B^{(m)} = \frac{1}{2}\,q^4 + \frac{1}{2}\,q^3 + \frac{5}{2}\,q^2 + 1$$
$$C^{(m)} = -\frac{1}{2}\,q^5 - \frac{1}{2}\,q^4 - \frac{3}{2}\,q^3 - q^2, \qquad K^{(m)} = -\frac{1}{4}, \qquad D^{(m)} = -\frac{3}{2}\,q + 1$$

$m = 6$:

$$A^{(m)} = -2\,q^3 - 6\,q + 4, \qquad B^{(m)} = -2\,q^4 - q^3 - 6\,q^2 + 1, \qquad C^{(m)} = -\frac{1}{2}\,q^5 - \frac{1}{2}\,q^4 - \frac{3}{2}\,q^3 - q^2$$
$$K^{(m)} = 4, \qquad D^{(m)} = \frac{3}{2}\,q + 1$$

$m = 7$:

$$A^{(m)} = \frac{1}{8}\,q^3 - \frac{1}{8}\,q + \frac{1}{4}, \qquad B^{(m)} = \frac{1}{2}\,q^4 + \frac{3}{2}\,q^2 + 1, \qquad C^{(m)} = \frac{1}{2}\,q^5 + \frac{3}{2}\,q^3 - q^2, \qquad K^{(m)} = \frac{1}{4}$$
$$D^{(m)} = \frac{1}{2}\,q + 1$$

$m = 8$:

$$A^{(m)} = 2, \qquad B^{(m)} = q^3 - 2\,q^2 + 1, \qquad C^{(m)} = -\frac{1}{2}\,q^5 + \frac{1}{2}\,q^3 - q^2, \qquad K^{(m)} = 2, \qquad D^{(m)} = 1$$

$m = 9$:

$$A^{(m)} = q^3, \qquad B^{(m)} = -q^3 + 2\,q^2 + 1, \qquad C^{(m)} = -q^2, \qquad K^{(m)} = q^3$$
$$D^{(m)} = -q^3 + 2\,q^2 + 1$$

$m = 10$:

$$A^{(m)} = 1, \qquad B^{(m)} = -q^3 + 2\,q^2 + 1, \qquad C^{(m)} = -q^5, \qquad K^{(m)} = 1, \qquad D^{(m)} = 1$$

## Appendix C. Base case $n = 3$

This appendix lists the 15 tuples for $m = 1, \ldots, 15$, computed by `verify_base.sage`. The period is 14 (from $m = 2$), and $m = 15$ maps back to $m = 2$. $m = 1$:

$$A^{(m)} = -q, \qquad B^{(m)} = -q^4 + 2\,q^3 + q^2 + 1, \qquad C^{(m)} = q^6, \qquad K^{(m)} = -q, \qquad D^{(m)} = q^2 + 1$$

$m = 2$:

$$A^{(m)} = q^4 - q^3 + 2\,q^2 - q + 2, \qquad B^{(m)} = q^4 - 2\,q^3 + q^2 + 1, \qquad C^{(m)} = -q^3, \qquad K^{(m)} = 2$$
$$D^{(m)} = \frac{1}{2}\,q + 1$$

$m = 3$:

$$A^{(m)} = -\frac{1}{8}\,q^4 + \frac{1}{8}\,q^3 - \frac{1}{4}\,q^2 + \frac{1}{8}\,q + \frac{1}{4}, \qquad B^{(m)} = \frac{1}{2}\,q^5 - \frac{1}{2}\,q^4 + 2\,q^3 + \frac{1}{2}\,q^2 + 1$$
$$C^{(m)} = -\frac{1}{2}\,q^6 + \frac{1}{2}\,q^5 - q^4 + \frac{1}{2}\,q^3 - q^2, \qquad K^{(m)} = \frac{1}{4}, \qquad D^{(m)} = -\frac{1}{2}\,q + 1$$

$m = 4$:

$$A^{(m)} = 2, \qquad B^{(m)} = -q^4 - 3\,q^2 + 1, \qquad C^{(m)} = \frac{1}{2}\,q^6 - \frac{1}{2}\,q^5 + q^4 - \frac{1}{2}\,q^3 - q^2, \qquad K^{(m)} = 2$$
$$D^{(m)} = 1$$

$m = 5$:

$$A^{(m)} = -q^4 + q^3 - 3\,q^2 + q - 1, \qquad B^{(m)} = q^4 + 3\,q^2 + 1, \qquad C^{(m)} = -q^2, \qquad K^{(m)} = -1$$
$$D^{(m)} = q + 1$$

$m = 6$:

$$A^{(m)} = -q^4 - q^2 - q, \qquad B^{(m)} = 2\,q^5 - q^4 + 4\,q^3 + q^2 + 1$$
$$C^{(m)} = -q^6 + q^5 - 3\,q^4 + q^3 - q^2, \qquad K^{(m)} = -q, \qquad D^{(m)} = 2\,q^2 - q + 1$$

$m = 7$:

$$A^{(m)} = -q^4 + q^3 - q^2, \qquad B^{(m)} = 2\,q^5 - q^4 + 2\,q^3 + q^2 + 1, \qquad C^{(m)} = -q^6 - q^4 - q^3$$
$$K^{(m)} = -q^2, \qquad D^{(m)} = 2\,q^3 + q^2 + q + 1$$

$m = 8$:

$$A^{(m)} = -q^4 - q^2 - q, \qquad B^{(m)} = 2\,q^5 - q^4 + 2\,q^3 + q^2 + 1, \qquad C^{(m)} = -q^6 + q^5 - q^4$$
$$K^{(m)} = -q, \qquad D^{(m)} = 2\,q^2 - q + 1$$

$m = 9$:

$$A^{(m)} = -q^4 + q^3 - 3\,q^2 + q - 1, \qquad B^{(m)} = 2\,q^5 - q^4 + 4\,q^3 + q^2 + 1, \qquad C^{(m)} = -q^6 - q^4 - q^3$$
$$K^{(m)} = -1, \qquad D^{(m)} = q + 1$$

$m = 10$:

$$A^{(m)} = -1, \qquad B^{(m)} = q^4 + 3\,q^2 + 1, \qquad C^{(m)} = -q^6 + q^5 - 3\,q^4 + q^3 - q^2, \qquad K^{(m)} = -1$$
$$D^{(m)} = 1$$

$m = 11$:

$$A^{(m)} = -q^4 + q^3 - 2\,q^2 + q + 2, \qquad B^{(m)} = -q^4 - 3\,q^2 + 1, \qquad C^{(m)} = -q^2, \qquad K^{(m)} = 2$$

$$D^{(m)} = -\frac{1}{2}\,q + 1$$

$m = 12$:

$$A^{(m)} = \frac{1}{8}\,q^4 - \frac{1}{8}\,q^3 + \frac{1}{4}\,q^2 - \frac{1}{8}\,q + \frac{1}{4}, \qquad B^{(m)} = \frac{1}{2}\,q^5 - \frac{1}{2}\,q^4 + 2\,q^3 + \frac{1}{2}\,q^2 + 1$$
$$C^{(m)} = \frac{1}{2}\,q^6 - \frac{1}{2}\,q^5 + q^4 - \frac{1}{2}\,q^3 - q^2, \qquad K^{(m)} = \frac{1}{4}, \qquad D^{(m)} = \frac{1}{2}\,q + 1$$

$m = 13$:

$$A^{(m)} = 2\,q, \qquad B^{(m)} = q^4 - 2\,q^3 + q^2 + 1, \qquad C^{(m)} = -\frac{1}{2}\,q^6 + \frac{1}{2}\,q^5 - q^4 + \frac{1}{2}\,q^3 - q^2$$
$$K^{(m)} = 2\,q, \qquad D^{(m)} = q^2 + 1$$

$m = 14$:

$$A^{(m)} = q^4, \qquad B^{(m)} = -q^4 + 2\,q^3 + q^2 + 1, \qquad C^{(m)} = -q^3, \qquad K^{(m)} = q^4$$
$$D^{(m)} = -q^4 + 2\,q^3 + q^2 + 1$$

$m = 15$:

$$A^{(m)} = q, \qquad B^{(m)} = -q^4 + 2\,q^3 + q^2 + 1, \qquad C^{(m)} = -q^6, \qquad K^{(m)} = q, \qquad D^{(m)} = q^2 + 1$$

## Appendix D. Pass 1

This appendix lists the tuples of pass 1, indexed by $m = 9j + r$ ($r \in \{1, \dots, 9\}$), starting from the initial tuple (19) at $m = 1$. The formulas hold for every $n \geq 4$, even or odd. Each generic row is written with the range on which its triple $(A, B, C)$ is admissible, i.e. polynomial with $B(0) = 1$, $C(0) = 0$.[1] The step from a row to the next is correct exactly where both rows are admissible (§3.4), which determines where pass 1 ends (§3.5). The rows $j = 0$ are used for every $n \geq 4$.

$m = 1$, $m = 9j + 1$, $j = 0$:

$$A^{(m)} = -q^{n-2}, \qquad B^{(m)} = -\frac{\big(q^2 - 3\,q + 1\big)q^n + Q}{q - 1}, \qquad C^{(m)} = q^{n+3}, \qquad K^{(m)} = -q^{n-2}$$
$$D^{(m)} = -\frac{Q - q^n}{q - 1}$$

$m = 9j + 1$, $j \geq 1$, $n - 2j \geq 0$:

$$A^{(m)} = -\frac{Q(q-2)^2 q^{4j} + Q(q-2)q^{2j}\,(1 + q^n) + q^{n+1}}{(q-1)^2 q^{2j}}, \qquad B^{(m)} = -\frac{Q\,\big(2(q-2)q^{2j} + q^n + 1\big)}{q - 1}$$
$$C^{(m)} = -Qq^{2\,j}, \qquad K^{(m)} = 2, \qquad D^{(m)} = -\frac{1}{2}\,q + 1$$

$m = 2$, $m = 9j + 2$, $j = 0$:

$$A^{(m)} = \frac{\big(q^2 - 2q + 2\big)\,q^n + Q(q-2)}{q - 1}, \qquad B^{(m)} = \frac{\big(q^2 - 3\,q + 3\big)q^n - Q}{q - 1}, \qquad C^{(m)} = -q^n$$
$$K^{(m)} = 2, \qquad D^{(m)} = \frac{1}{2}\,q + 1$$

$m = 9j + 2$, $j \geq 1$, $n - 2j \geq 1$:

[1] This is stronger than polynomiality: rows 4, 8, 9 of pass 1 and rows 3, 8, 9 of pass 2 are still polynomial at $n - 2j = p_r - 1$, but there $C(0) \neq 0$ (rows 4, 8 of pass 1 have $C = -1$, row 9 has $C(0) = 1$; rows 3, 8 of pass 2 have $C = -Q$, row 9 has $C(0) = -1$).

$$A^{(m)} = \frac{Qq(q-2)q^{2j}\,(1+q^n) + Qq^3q^{4j} + (q-2)^2q^n}{8q(q-1)^2q^{2j}}$$

$$B^{(m)} = \frac{Qq^2(q-2)q^{4j} + Q\left(q^2-2q+2\right)q^{2j}\,(1+q^n) + q(q-2)q^n}{2(q-1)^2q^{2j}}$$

$$C^{(m)} = \frac{Qq^2(q-2)^2q^{4j} + Qq^2(q-2)q^{2j}\,(1+q^n) + q^3q^n}{2(q-1)^2q^{2j}}, \qquad K^{(m)} = -\frac{1}{4}, \qquad D^{(m)} = -\frac{1}{2}\,q+1$$

$m = 3$, $m = 9j+3$, $j = 0$:

$$A^{(m)} = -\frac{Qq(q+2) + (q-2)\left(q^2+2\right)q^n}{8q(q-1)}, \qquad B^{(m)} = \frac{Q\left(q^2-2\right) + \left(q^3-2q^2+4q-2\right)q^n}{2(q-1)}$$

$$C^{(m)} = -\frac{q^2\left(Q(q-2) + \left(q^2-2q+2\right)q^n\right)}{2(q-1)}, \qquad K^{(m)} = \frac{1}{4}, \qquad D^{(m)} = -\frac{1}{2}\,q+1$$

$m = 9j+3$, $j \geq 1$, $n-2j \geq 3$:

$$A^{(m)} = 2\,q^{-2\,j+n-3}, \qquad B^{(m)} = -\frac{Qq\,q^{2j}\,(1+q^n) + 2(q-2)q^n}{q(q-1)q^{2j}}$$

$$C^{(m)} = \frac{Qq^2(q-2)q^{2j}\,(1+q^n) + Qq^4q^{4j} + q(q-2)^2q^n}{2(q-1)^2q^{2j}}, \qquad K^{(m)} = 2\,q^{-2\,j+n-3}$$

$$D^{(m)} = -\frac{Qq^{2\,j+1} - q^n}{(q^2-q)q^{2\,j}}$$

$m = 4$, $m = 9j+4$, $j = 0$:

$$A^{(m)} = 2\,q^{n-3}, \qquad B^{(m)} = -\frac{\left(q^3-q^2+3\,q-4\right)q^n + Qq}{q^2-q}$$

$$C^{(m)} = \frac{q\left(Qq(q+2) + (q-2)\left(q^2+2\right)q^n\right)}{2(q-1)}, \qquad K^{(m)} = 2\,q^{n-3}, \qquad D^{(m)} = -\frac{Qq-q^n}{q^2-q}$$

$m = 9j+4$, $j \geq 1$, $n-2j \geq 2$:

$$A^{(m)} = \frac{Qq^2(q-1)q^{4j} + Qq\,q^{2j}\,(1-q^n) - (q-1)q^n}{q(q-1)q^{2j}}, \qquad B^{(m)} = -\frac{Qq\,q^{2j}\,(1-q^n) - 2(q-1)q^n}{q(q-1)q^{2j}}$$

$$C^{(m)} = -q^{-2\,j+n-1}, \qquad K^{(m)} = -1, \qquad D^{(m)} = 1$$

$m = 5$, $m = 9j+5$, $j = 0$:

$$A^{(m)} = \frac{Q^2q - \left(q^3-q^2+2\,q-1\right)q^n}{q^2-q}, \qquad B^{(m)} = \frac{\left(q^3-q^2+3\,q-2\right)q^n - Qq}{q^2-q}$$

$$C^{(m)} = -q^{n-1}, \qquad K^{(m)} = -1, \qquad D^{(m)} = q+1$$

$m = 9j+5$, $j \geq 1$, $n-2j \geq 0$:

$$A^{(m)} = Qq^{2\,j-1}, \qquad B^{(m)} = -\frac{Q\left(2q(q-1)q^{2j} - q^n + 1\right)}{q-1}$$

$$C^{(m)} = \frac{Qq^3(q-1)q^{4j} + Qq^2q^{2j}\,(1-q^n) - q(q-1)q^n}{(q-1)q^{2j}}, \qquad K^{(m)} = q^{2\,j-1}, \qquad D^{(m)} = \frac{q^{2\,j+1}-1}{q-1}$$

$m = 6$, $m = 9j+6$, $j = 0$:

$$A^{(m)} = \frac{Qq(q+1) - \left(q^2+1\right)q^n}{q-1}, \qquad B^{(m)} = -\frac{Q\left(2q^3+1\right) - \left(2q^3-q^2+3q-1\right)q^n}{q-1}$$

$$C^{(m)} = \frac{q\left(Q^2 q - \left(q^3 - q^2 + 2q - 1\right) q^n\right)}{q-1}, \qquad K^{(m)} = -q, \qquad D^{(m)} = Q$$

$m = 9j + 6$, $j \geq 1$, $n - 2j \geq 0$:

$$A^{(m)} = -\frac{Qq^3 q^{4j} - Qq\, q^{2j}\,(1+q^n) + q^n}{(q-1)^2 q^{2j}}, \qquad B^{(m)} = \frac{Q\left(2q^2 q^{2j} - q^n - 1\right)}{q-1}, \qquad C^{(m)} = -Qq^{2\,j+1}$$

$$K^{(m)} = q, \qquad D^{(m)} = -q + 1$$

$m = 9j + 7$, $j \geq 0$, $n - 2j \geq 4$:

$$A^{(m)} = q^{-2\,j+n-4}, \qquad B^{(m)} = -\frac{Qq\, q^{2j}\,(1+q^n) - 2q^n}{q(q-1)q^{2j}}$$

$$C^{(m)} = \frac{Qq^5 q^{4j} - Qq^3 q^{2j}\,(1+q^n) + q^2 q^n}{(q-1)^2 q^{2j}}, \qquad K^{(m)} = q^{-2\,j+n-4}, \qquad D^{(m)} = -\frac{Qq^{2\,j+2} - q^n}{(q^3 - q^2)q^{2\,j}}$$

$m = 9j + 8$, $j \geq 0$, $n - 2j \geq 3$:

$$A^{(m)} = \frac{Qq^4(q-1)q^{4j} - Qq^2 q^{2j}\,(1-q^n) - (q-1)q^n}{q^2(q-1)q^{2j}}$$

$$B^{(m)} = -\frac{Qq^2 q^{2j}\,(1-q^n) + 2(q-1)q^n}{q^2(q-1)q^{2j}}, \qquad C^{(m)} = -q^{-2\,j+n-2}, \qquad K^{(m)} = 1, \qquad D^{(m)} = 1$$

$m = 9j + 9$, $j \geq 0$, $n - 2j \geq 1$:

$$A^{(m)} = -Qq^{2\,j}, \qquad B^{(m)} = \frac{Q\left(2q^2(q-1)q^{2j} + q^n - 1\right)}{q-1}$$

$$C^{(m)} = -\frac{Qq^4(q-1)q^{4j} - Qq^2 q^{2j}\,(1-q^n) - (q-1)q^n}{(q-1)q^{2j}}, \qquad K^{(m)} = -q^{2\,j}$$

$$D^{(m)} = \frac{q^{2\,j+2} - 1}{q-1}$$

## Appendix E. Pass 2

This appendix lists the tuples of pass 2, indexed by $m = 9j + r$ ($r \in \{1, \ldots, 9\}$) and written in terms of $q^{2j}$ and $q^n$. Here $m$ is an index local to pass 2: where pass 2 sits in the orbit, and at which $j$ its rows are read, is described in §3.5. The formulas hold for every $n \geq 4$, even or odd. As in Appendix D, each row is written with the range on which its triple $(A, B, C)$ is admissible. A bound $j \geq \frac{1}{2}$ means $j \geq 1$ when the rows are read at integer $j$ (even $n$).

$m = 9j + 1$, $j \geq 1$, $n - 2j \geq 0$:

$$A^{(m)} = \frac{Qq\left(q^{2j} + (q^{2j} - 1)q^n\right) - q^{4j+2}}{(q-1)^2 q^{2j}}, \qquad B^{(m)} = -\frac{Q(q^n + 1) - 2q^{2j+1}}{q-1}, \qquad C^{(m)} = -q^{2\,j}$$

$$K^{(m)} = q, \qquad D^{(m)} = -q + 1$$

$m = 9j + 2$, $j \geq \frac{1}{2}$, $n - 2j \geq 3$:

$$A^{(m)} = Qq^{-2\,j+n-3}, \qquad B^{(m)} = -\frac{Q\left(q^{2j} + (q^{2j} - 2)q^n\right)}{(q-1)q^{2j}}$$

$$C^{(m)} = -\frac{Qq^3\left(q^{2j} + (q^{2j} - 1)q^n\right) - q^{4j+4}}{(q-1)^2 q^{2j}}, \qquad K^{(m)} = q^{-2\,j+n-3}, \qquad D^{(m)} = -\frac{q^{2\,j+1} - q^n}{(q^2 - q)q^{2\,j}}$$

$m = 9j + 3$, $j \geq \frac{1}{2}$, $n - 2j \geq 2$:

$$A^{(m)} = \frac{q^2(q-1)q^{4j} + Q\left((q^{2j+1} - q + 1)q^n - q^{2j+1}\right)}{(q^2-q)q^{2j}}$$
$$B^{(m)} = -\frac{Q\left(q^{2j+1} - (q^{2j+1} - 2q + 2)q^n\right)}{(q^2-q)q^{2j}}, \qquad C^{(m)} = -Qq^{-2\,j+n-1}, \qquad K^{(m)} = 1$$
$$D^{(m)} = 1$$

$m = 9j+4$, $j \geq \frac{1}{2}$, $n - 2j \geq 0$:

$$A^{(m)} = -q^{2\,j-1}, \qquad B^{(m)} = \frac{2q(q-1)q^{2j} + Q(q^n - 1)}{q-1}$$
$$C^{(m)} = -\frac{(q^4 - q^3)q^{4j} - Q\left(q^{2j+2} + (q^2 - q - q^{2j+2})q^n\right)}{(q-1)q^{2j}}, \qquad K^{(m)} = -q^{2\,j-1}$$
$$D^{(m)} = -\frac{Q - q^{2\,j+1}}{q-1}$$

$m = 9j+5$, $j \geq \frac{1}{2}$, $n - 2j \geq 0$:

$$A^{(m)} = -\frac{q(q-2)^2q^{4j} + Q(q-2)q^{2j}(1+q^n) + Qq^n}{(q-1)^2q^{2j}}, \qquad B^{(m)} = -\frac{2q(q-2)q^{2j} + Q(q^n+1)}{q-1}$$
$$C^{(m)} = -q^{2\,j+1}, \qquad K^{(m)} = 2, \qquad D^{(m)} = -\frac{1}{2}\,q + 1$$

$m = 9j+6$, $j \geq 0$, $n - 2j \geq 2$:

$$A^{(m)} = \frac{Q(q-2)\left(q^{2j+2} + (q^{2j+2} + q - 2)q^n\right) + q^{4j+5}}{8q^2(q-1)^2q^{2j}}$$
$$B^{(m)} = \frac{(q^4 - 2q^3)q^{4j} + Q(q^2 - 2q + 2)q^{2j}(1+q^n) + Q(q-2)q^n}{2(q-1)^2q^{2j}}$$
$$C^{(m)} = \frac{(q^5 - 4q^4 + 4q^3)q^{4j} + Qq^2(q-2)q^{2j}(1+q^n) + Qq^2q^n}{2(q-1)^2q^{2j}}, \qquad K^{(m)} = -\frac{1}{4}$$
$$D^{(m)} = -\frac{1}{2}\,q + 1$$

$m = 9j+7$, $j \geq 0$, $n - 2j \geq 4$:

$$A^{(m)} = 2\,Qq^{-2\,j+n-4}, \qquad B^{(m)} = -\frac{Q\left(q^{2j+2} + (q^{2j+2} + 2q - 4)q^n\right)}{(q^3-q^2)q^{2j}}$$
$$C^{(m)} = \frac{Q(q-2)\left(q^{2j+2} + (q^{2j+2} + q - 2)q^n\right) + q^{4j+5}}{2(q-1)^2q^{2j}}, \qquad K^{(m)} = 2\,q^{-2\,j+n-4}$$
$$D^{(m)} = -\frac{q^{2\,j+2} - q^n}{(q^3-q^2)q^{2\,j}}$$

$m = 9j+8$, $j \geq 0$, $n - 2j \geq 3$:

$$A^{(m)} = \frac{(q^5 - q^4)q^{4j} + Q\left(q^{2j+2} - (q^{2j+2} + q - 1)q^n\right)}{(q^3-q^2)q^{2j}}$$
$$B^{(m)} = -\frac{Q\left(q^{2j+2} - (q^{2j+2} + 2q - 2)q^n\right)}{(q^3-q^2)q^{2j}}, \qquad C^{(m)} = -Qq^{-2\,j+n-2}, \qquad K^{(m)} = -1$$
$$D^{(m)} = 1$$

$m = 9j+9$, $j \geq 0$, $n - 2j \geq 1$:

$$A^{(m)} = q^{2\,j}, \qquad B^{(m)} = -\frac{2q^2(q-1)q^{2j} - Q(q^n-1)}{q-1}$$
$$C^{(m)} = \frac{(q^5-q^4)q^{4j} + Q\left(q^{2j+2} - (q^{2j+2}+q-1)q^n\right)}{(q-1)q^{2j}}, \qquad K^{(m)} = q^{2\,j}, \qquad D^{(m)} = -\frac{Q-q^{2\,j+2}}{q-1}$$

## Appendix F. Bridge 1

This appendix lists the eight tuples b1, ..., b8 of bridge 1. They do not involve $j$, and the formulas hold for every $n \geq 4$, even or odd.

b1:

$$A^{(\mathrm{b1})} = -\frac{Qq(q+1) - (q^2+1)q^n}{q-1}, \qquad B^{(\mathrm{b1})} = -\frac{\left(q^3-q^2+q-2\right)q^n + Qq}{q^2-q}, \qquad C^{(\mathrm{b1})} = -q^{n-2}$$
$$K^{(\mathrm{b1})} = q, \qquad D^{(\mathrm{b1})} = Q$$

b2:

$$A^{(\mathrm{b2})} = \frac{Q^2q - \left(q^3-q^2+2\,q-1\right)q^n}{q^2-q}, \qquad B^{(\mathrm{b2})} = -\frac{Q(2q^3+1) - (2\,q^3-q^2+3\,q-1)q^n}{q-1}$$
$$C^{(\mathrm{b2})} = \frac{Qq^3(q+1) - q^2(q^2+1)q^n}{q-1}, \qquad K^{(\mathrm{b2})} = -1, \qquad D^{(\mathrm{b2})} = q+1$$

b3:

$$A^{(\mathrm{b3})} = -q^{n-3}, \qquad B^{(\mathrm{b3})} = \frac{\left(q^3-q^2+3\,q-2\right)q^n - Qq}{q^2-q}$$
$$C^{(\mathrm{b3})} = \frac{Q^2q^2 - \left(q^4-q^3+2\,q^2-q\right)q^n}{q-1}, \qquad K^{(\mathrm{b3})} = -q^{n-3}, \qquad D^{(\mathrm{b3})} = -\frac{Qq-q^n}{q^2-q}$$

b4:

$$A^{(\mathrm{b4})} = -\frac{Qq(q+2) + (q-2)(q^2+2)q^n}{q^2-q}, \qquad B^{(\mathrm{b4})} = -\frac{\left(q^3-q^2+3\,q-4\right)q^n + Qq}{q^2-q}$$
$$C^{(\mathrm{b4})} = -q^{n-1}, \qquad K^{(\mathrm{b4})} = 2, \qquad D^{(\mathrm{b4})} = -\frac{1}{2}\,q+1$$

b5:

$$A^{(\mathrm{b5})} = \frac{(q^2-2\,q+2)q^n + Q(q-2)}{8\,(q-1)}, \qquad B^{(\mathrm{b5})} = \frac{Q(q^2-2) + (q^3-2\,q^2+4\,q-2)q^n}{2\,(q-1)}$$
$$C^{(\mathrm{b5})} = \frac{Qq^2(q+2) + q(q-2)(q^2+2)q^n}{2\,(q-1)}, \qquad K^{(\mathrm{b5})} = \frac{1}{4}, \qquad D^{(\mathrm{b5})} = \frac{1}{2}\,q+1$$

b6:

$$A^{(\mathrm{b6})} = 2\,q^{n-2}, \qquad B^{(\mathrm{b6})} = \frac{\left(q^2-3\,q+3\right)q^n - Q}{q-1}, \qquad C^{(\mathrm{b6})} = -\frac{q^2\left(Q(q-2) + (q^2-2q+2)q^n\right)}{2\,(q-1)}$$
$$K^{(\mathrm{b6})} = 2\,q^{n-2}, \qquad D^{(\mathrm{b6})} = -\frac{Q-q^n}{q-1}$$

b7:

$$A^{(\mathrm{b7})} = q^{n+1}, \qquad B^{(\mathrm{b7})} = -\frac{\left(q^2-3\,q+1\right)q^n + Q}{q-1}, \qquad C^{(\mathrm{b7})} = -q^n, \qquad K^{(\mathrm{b7})} = q^{n+1}$$
$$D^{(\mathrm{b7})} = -\frac{\left(q^2-3\,q+1\right)q^n + Q}{q-1}$$

b8:

$$A^{(\mathrm{b8})} = q^{n-2}, \qquad B^{(\mathrm{b8})} = -\frac{\left(q^2-3\,q+1\right)q^n+Q}{q-1}, \qquad C^{(\mathrm{b8})} = -q^{n+3}, \qquad K^{(\mathrm{b8})} = q^{n-2}$$
$$D^{(\mathrm{b8})} = -\frac{Q-q^n}{q-1}$$

## Appendix G. Bridge 2

This appendix lists the five tuples $\mathrm{c1},\dots,\mathrm{c5}$ of bridge 2. They do not involve $j$, and the formulas hold for every $n \geq 4$, even or odd.

c1:

$$A^{(\mathrm{c1})} = \frac{Qq^n+(2q-1)(q-2)}{8\,(q-1)}, \qquad B^{(\mathrm{c1})} = \frac{2\,q^3+Qq^{n+1}-3\,q^2+2\,q-2}{2\,(q-1)}$$
$$C^{(\mathrm{c1})} = \frac{q^2(2q^2-q+2)+Q(q-2)(q+2)q^n}{2\,(q-1)}, \qquad K^{(\mathrm{c1})} = -\frac{1}{4}, \qquad D^{(\mathrm{c1})} = \frac{3}{2}\,q+1$$

c2:

$$A^{(\mathrm{c2})} = \frac{2\left(2q(q^2-q-1)+Q(q+1)q^n\right)}{q(q-1)}, \qquad B^{(\mathrm{c2})} = -\frac{4q^3-7q^2-q+1+Q(2q+1)q^n}{q-1}$$
$$C^{(\mathrm{c2})} = \frac{q^2\left((2q-1)(q-2)+Qq^n\right)}{2\,(q-1)}, \qquad K^{(\mathrm{c2})} = 4, \qquad D^{(\mathrm{c2})} = -2\,q+1$$

c3:

$$A^{(\mathrm{c3})} = \frac{1}{2}\,Qq^{n-3}, \qquad B^{(\mathrm{c3})} = \frac{Qq^n-Qq}{q^2-q}, \qquad C^{(\mathrm{c3})} = -\frac{2q^2(q^2-q-1)+Qq(q+1)q^n}{2\,(q-1)}$$
$$K^{(\mathrm{c3})} = \frac{1}{2}\,q^{n-3}, \qquad D^{(\mathrm{c3})} = \frac{q^n-q}{q^2-q}$$

c4:

$$A^{(\mathrm{c4})} = \frac{2q(q^2-q-1)+Q(q+1)q^n}{4\,(q^2-q)}, \qquad B^{(\mathrm{c4})} = \frac{Qq^n-Qq}{q^2-q}, \qquad C^{(\mathrm{c4})} = -Qq^{n-1}$$
$$K^{(\mathrm{c4})} = \frac{1}{2}, \qquad D^{(\mathrm{c4})} = -2\,q+1$$

c5:

$$A^{(\mathrm{c5})} = -\frac{2\,(Qq^n+(2q-1)(q-2))}{q-1}, \qquad B^{(\mathrm{c5})} = -\frac{4q^3-7q^2-q+1+Q(2q+1)q^n}{q-1}$$
$$C^{(\mathrm{c5})} = -\frac{2q^2(q^2-q-1)+Qq(q+1)q^n}{2\,(q-1)}, \qquad K^{(\mathrm{c5})} = 4, \qquad D^{(\mathrm{c5})} = \frac{3}{2}\,q+1$$

## Appendix H. Bridge 3

This appendix lists the three tuples $\mathrm{d1},\mathrm{d2},\mathrm{d3}$ of bridge 3. They do not involve $j$, and the formulas hold for every $n \geq 4$, even or odd.

d1:

$$A^{(\mathrm{d1})} = \frac{Qq^n-q}{q-1}, \qquad B^{(\mathrm{d1})} = -\frac{Q\left((q-2)q^n+q\right)}{q^2-q}, \qquad C^{(\mathrm{d1})} = -Qq^{n-2}, \qquad K^{(\mathrm{d1})} = q$$
$$D^{(\mathrm{d1})} = Q$$

d2:

$$A^{(\mathrm{d2})}=-Qq^{n-1},\qquad B^{(\mathrm{d2})}=\frac{Q\big((2q-1)q^n-1\big)}{q-1},\qquad C^{(\mathrm{d2})}=\frac{q^3-Qq^{n+2}}{q-1},\qquad K^{(\mathrm{d2})}=-q^{n-1}$$
$$D^{(\mathrm{d2})}=\frac{(2\,q-1)q^n-1}{q-1}$$

d3:

$$A^{(\mathrm{d3})}=-\frac{Qq^n-q}{q-1},\qquad B^{(\mathrm{d3})}=\frac{Q\big((2q-1)q^n-1\big)}{q-1},\qquad C^{(\mathrm{d3})}=-Qq^{n+1},\qquad K^{(\mathrm{d3})}=-q$$
$$D^{(\mathrm{d3})}=Q$$

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